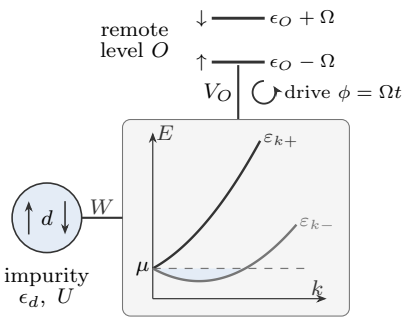
(a) Driven impurity model



radial SOC bath, full k

drive $\Rightarrow$ coherent spin-odd coupling, $\mathrm{Re}\,\Sigma_z$
SOC, $\rho_- \neq \rho_+ \Rightarrow$ relative decay, $\mathrm{Im}\,\Sigma_x$
the EP needs both

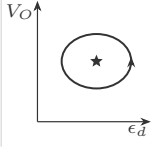
(b) Resonance EP at $U = 0$





one control loop: $z_1 \leftrightarrow z_2$

$F(z) = \det[(z - \epsilon_d)I - \Sigma_d^{\mathrm{II}}(z)]$



EP: $F = \partial_z F = 0$
loop: $\mathcal{D}_{\mathrm{loc}}$ winds ± 1
family in λ (SOC)

(c) Interaction evidence

Nearly empty: $\langle n_d \rangle \leq 0.035$

finite-chain embedding: $U = \bar{\Gamma}_0$
continuum KR saddle: $U = 80\bar{\Gamma}_0$
pole exchange checked in each approximation

Charge fluctuating: $\langle n_d \rangle \simeq 1$

exactly filling-one KR branches have folds
nearby EP: $U = 0.0032$, $\langle n_d \rangle = 0.986$
$D_d = 0.213$: a local moment has not formed

Local-moment EP: unresolved

exact interacting continuum convergence remains a separate calculation